

Subject Description Form

Subject Code	ITC3203T
Subject Title	Function-Focus Fibrous Products and Technology
Credit Value	3
Level	3
Pre-requisite(s) / <Co-requisite> / (Exclusion)	Nil
Objectives	The subject develops understanding of the concepts, principles, materials, structures and properties as well as manufacturing processes for typical function-focus fibrous products used (FFFPs) in personal protection, hygiene and healthcare, transportation and construction. It identifies the performance requirements of FFFPs and their evaluation technologies. It also assesses the broad range of latest developments critically from the product development and market perspectives.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - understand and assess technologies involved in FFFPs including their functions, technical specifications, materials, manufacturing processes, as well as advantages and disadvantages; - describe the development trends in technologies for FFFPs and their industrial applications; - cooperate efficiently in a team to achieve goals; - develop critical and creative thinking; - develop the self-learning ability to adapt new technologies and update their knowledge; - develop the ability to source, select and integrate information.
Subject Synopsis/ Indicative Syllabus	(I) Concept and Classification of Function-Focus Fibrous Products Definition. Historical and current developments. Market needs. Materials. Design, Manufacturing. Certification. Supplier chains. Intellectual properties. (II) Product Design Conceptualization and Design Analysis Creativity. Brainstorming. Decision making. Concept optimization. Decision theory. State of nature. Payoff table. Decision tree. Bayes' theorem. Mathematical modeling. Empirical modeling. Failure analysis. Reliability and survival analysis.

	In the continuous assessment, by group projects and individual laboratory reports, focus will be made on all rounded education and self-learning ability. In addition to professional knowledge, the importance of team work, critical and creative thinking, the ability to adapt new technologies and update knowledge by self-learning, as well as the ability to source, select and integrate information will be stressed. For example, interaction among group members will be a very good way for enhancing communication skills, promoting team work spirit, leadership as well as entrepreneurship. In order to gain systematic technical knowledge/skills, students will also be given some questions as assignments. In the examination, students' learning performance in terms of defining concepts, explaining principles and describing and analysing application examples of technical textile products and related manufacturing processes will be assessed.	
Student Study Effort Expected	Class contact:	
	• Lecture	20 Hrs.
	• Laboratory	12 Hrs.
	• Seminars	6 Hrs
	Other student study effort:	
	• Assignments	25 Hrs.
	• Self study	42 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	<u>Recommended Textbooks</u> <u>Essentials</u> Cherif, C. (2016), <i>Textile materials for lightweight constructions: technologies - methods - materials – properties</i>. Springer. (e-version available) El Mogahzy, Y. (2009), <i>Engineering textiles</i>. Elsevier. (e-version available) Horrocks, A. R. and Anand. S. C. (2016), <i>Handbook of technical textiles</i>. Elsevier. <u>Supplementary</u> Alagirusamy, R. and Das, A. (2010), <i>Technical textile yarns industrial and medical applications</i>. Woodhead Publishing Ltd, Cambridge; CRC Press, US (e-version available) Chou, T. W. and Ko, F. K. (1989), <i>Textile Structural Composites</i>, Elsevier. Gutowski, T. G. (1997), <i>Advanced composites manufacturing</i>. John Wiley & Sons.	

	<u>Periodicals and Websites</u> Textile Research Journal High Performance Textiles Textile Outlook International Medical textiles
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